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Section: 7MA01

Course: ARTI 403 – Image Processing

Lab 9: Morphological Operations

Task 1: Apply morphological operations

In [4]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt

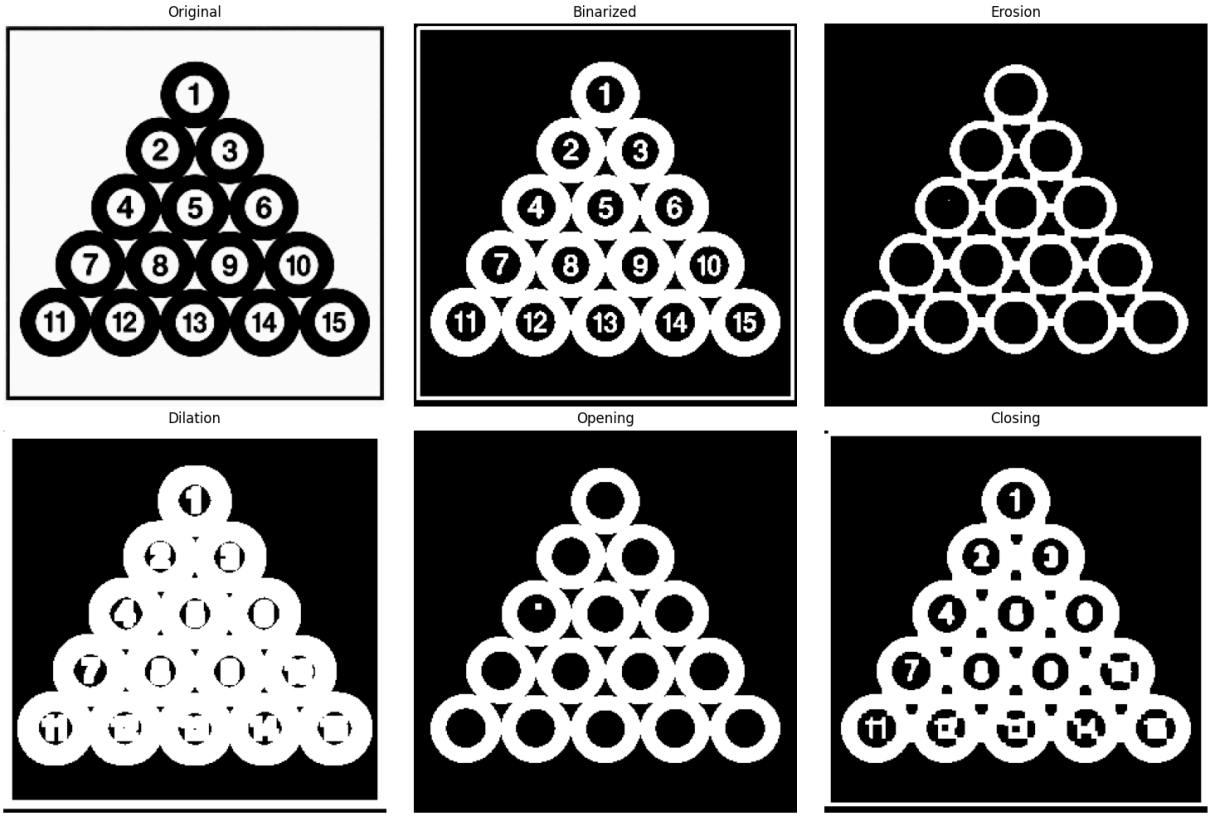
img = cv2.imread('Balls.png', 0)
img = cv2.resize(img,(300,300))
# Binarize image (Inverse so objects are white)
_, binary = cv2.threshold(img, 0, 255, cv2.THRESH_BINARY_INV + cv2.THRESH_OTSU)

kernel = np.ones((5, 5), np.uint8)

eroded_img = cv2.erode(binary, kernel, iterations=1)
dilated_img = cv2.dilate(binary, kernel, iterations=1)
opened_img = cv2.morphologyEx(binary, cv2.MORPH_OPEN, kernel)
closed_img = cv2.morphologyEx(binary, cv2.MORPH_CLOSE, kernel)

plt.figure(figsize=(15, 10))
plt.subplot(2, 3, 1), plt.imshow(img, cmap='gray'), plt.title('Original'), plt.axis
plt.subplot(2, 3, 2), plt.imshow(binary, cmap='gray'), plt.title('Binarized'), plt.
plt.subplot(2, 3, 3), plt.imshow(eroded_img, cmap='gray'), plt.title('Erosion'), pl
plt.subplot(2, 3, 4), plt.imshow(dilated_img, cmap='gray'), plt.title('Dilation'),
plt.subplot(2, 3, 5), plt.imshow(opened_img, cmap='gray'), plt.title('Opening'), pl
plt.subplot(2, 3, 6), plt.imshow(closed_img, cmap='gray'), plt.title('Closing'), pl

plt.tight_layout()
plt.show()
```



Task 2: Experiment with structuring elements

In [5]:

```
shapes = {
    'Square': cv2.MORPH_RECT,
    'Cross': cv2.MORPH_CROSS,
    'Ellipse': cv2.MORPH_ELLIPSE
}
sizes = [3, 5, 7]

fig_erode, axes_erode = plt.subplots(len(sizes), len(shapes), figsize=(12, 12))
fig_erode.suptitle('Erosion', fontsize=16)

fig_dilate, axes_dilate = plt.subplots(len(sizes), len(shapes), figsize=(12, 12))
fig_dilate.suptitle('Dilation', fontsize=16)

for i, size in enumerate(sizes):
    for j, (shape_name, shape_type) in enumerate(shapes.items()):

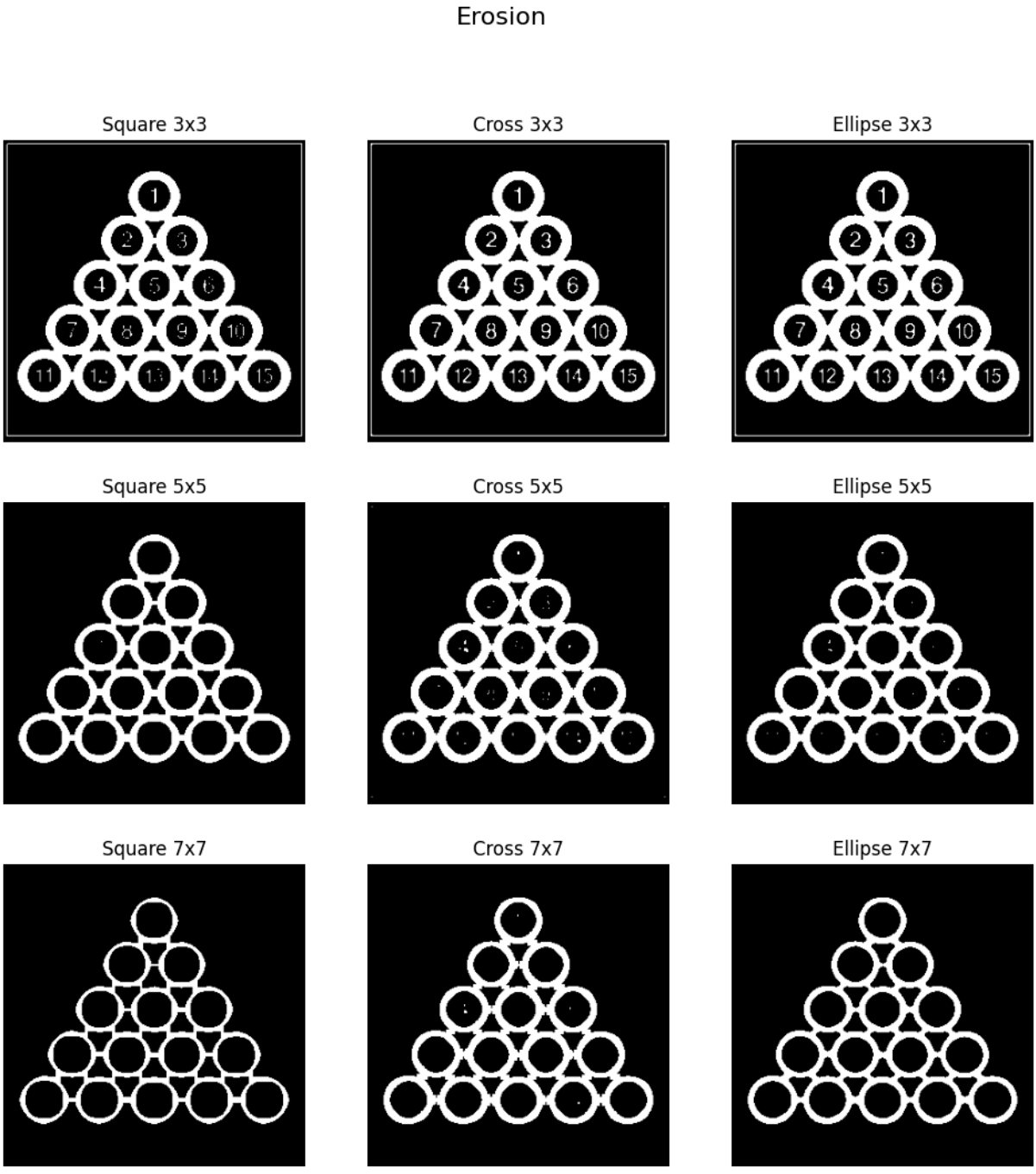
        kernel = cv2.getStructuringElement(shape_type, (size, size))

        eroded = cv2.erode(binary, kernel, iterations=1)
        dilated = cv2.dilate(binary, kernel, iterations=1)

        axes_erode[i, j].imshow(eroded, cmap='gray')
        axes_erode[i, j].set_title(f'{shape_name} {size}x{size}')
        axes_erode[i, j].axis('off')

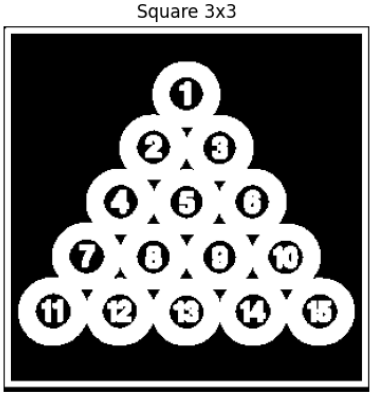
        axes_dilate[i, j].imshow(dilated, cmap='gray')
        axes_dilate[i, j].set_title(f'{shape_name} {size}x{size}')
        axes_dilate[i, j].axis('off')
```

```
plt.tight_layout()
plt.show()
```

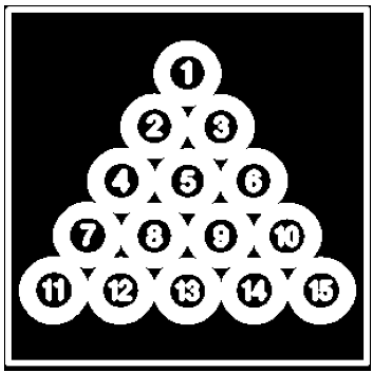


Dilation

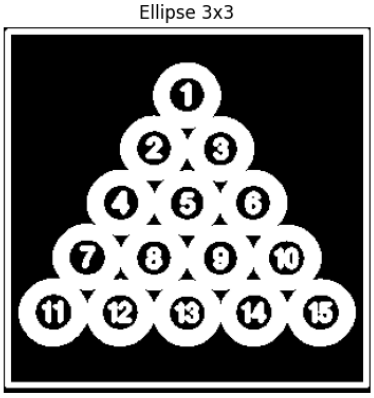
Square 3x3



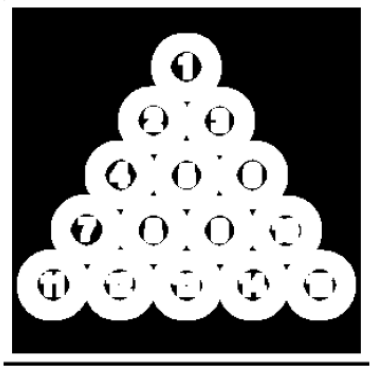
Cross 3x3



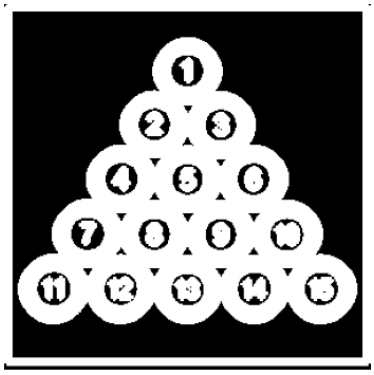
Ellipse 3x3



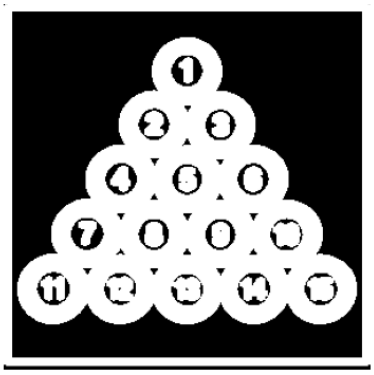
Square 5x5



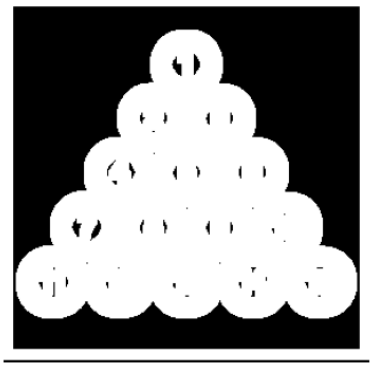
Cross 5x5



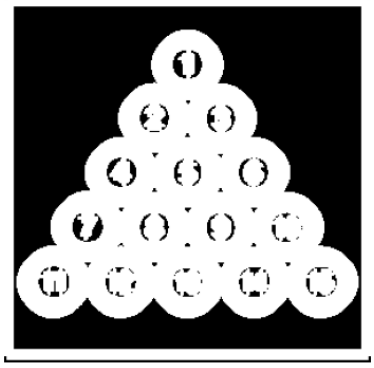
Ellipse 5x5



Square 7x7



Cross 7x7



Ellipse 7x7

